Supplement: A Change of Core: Revisiting ESP32 Benchmarks on ESP32-C6

1 Reader map

This supplement accompanies editorial revision 1.0.1 and analysis archive v1.0.1, doi:10.5281/zenodo.23493672. Package paths below are relative to the archive root. `data/local_timings.csv` contains 21,600 accepted timing records, not the complete raw console evidence. `source_lineage.csv` and `manifest.csv` record source and package hashes. Run `analysis.py`, then `matlab/run_publication_analysis.m` to recompute timing results and all four publication figures. The Python script also regenerates result tables from timing summaries and imported static metrics. The complete private archive separately retains protocols, inputs, raw captures, exact builds, and disassemblies. Those files are not supplied or promised on request while redistribution review is unresolved. Numerical correctness and emitted instructions cannot be independently re-audited from this package.

2 Historical-contract limitations

Area	Recovered issue	Treatment
Warm-up	Published prose and source behavior are not fully consistent.	One unmeasured warm-up is applied uniformly to new C6 implementations; historical values remain untouched.
MicroPython boundary	Slicing and preparation differ among historical implementations.	All new preparation, allocation, GC, and validation are outside the timed kernel.
Launcher iterations	Historical launcher behavior is not recoverable uniformly.	New C6 uses 100 accepted samples per size/language after correctness and pilot gates.
Validation	Historical validation coverage is incomplete.	Independent host/MATLAB oracles gate every new workload and language before timing.
Raw samples	Plauska reports aggregates rather than compatible per-invocation rows.	Published means remain aggregate evidence and are never expanded into invented samples.

3 Artifact interpretation

Exact C6 measured-build evidence is distinguished from matched reconstructed objects. The reconstruction fixes matched source and declared policies but not compiler backend, ABI, or implementation. Static instruction counts are not dynamic counts. The accepted Mode A Rust CRC ELF is the single disclosed preservation exception; its generated-code analysis is reconstruction only. The study supports claims about the observed board/build combinations, not a pure ISA treatment, population hardware effect, energy, dynamic instruction count, CPI, or cache behavior. Hardware SHA is a complete peripheral/API path; Rust FFT is a whole generated-kernel transformation.

4 Claim-to-evidence map

Claim layer	Package/private evidence and integrity locator	Command or access boundary	Permitted use
Mode A timing	Package: `data/local_timings.csv`; hashes in `manifest.csv`. Private: original raw-bundle manifests.	`python analysis.py`	One-board observation
Historical timing	Package: cited aggregates in `data/historical_means.csv`; external locators and hashes in `data/baselines/plauska_sources_v1.csv`. Source PDF/CSV not included.	`python analysis.py`; external source check remains separate	Descriptive comparison

Claim layer	Package/private evidence and integrity locator	Command or access boundary	Permitted use
Exact C6 code	Private: native artifacts under `artifacts/builds/`. Package: full retained hashes/status in `data/retained_artifact_hashes.csv`, including the Rust CRC omission.	Binaries/disassembly excluded; hashes identify artifacts but cannot supply a re-audit	Static measured-build evidence
Matched targets	Private: manifests under `artifacts/static/` and `artifacts/assembly/`.	Reconstruction sources/objects excluded	Mechanism only
Mode B	Package: `data/local_timings.csv`; derived `data/processed/mode_b_summary.csv`. Exact profile builds retained privately.	`python analysis.py`	Declared profile only
FFT transformation table	Package: imported numeric codegen metrics under `data/processed/`; source hashes in `source_lineage.csv`.	`python analysis.py`; does not regenerate metrics	Derived table only
Figures	Package: `paper/figures/`; hashes in `manifest.csv`.	`matlab/run_publication_analysis.m`	Derived display
Claims/citations	Package: `paper/CLAIM_AUDIT_2.md` and `paper/references.bib`; private full verification ledger.	`python verify_package.py`; not a new primary-source audit	Stated evidence boundaries

5 Complete historical/local matrix

The main paper uses a compact local-only headline table. This full matrix applies the same historical evidence field to every language and retains the published/local boundary.

Workload	Language	Local C6 median	Published mean	Published/local mean
		Time (μs)		
CRC-32	C/C++	104.000	117.630	1.134
CRC-32	Rust	72.000	113.750	1.578
CRC-32	MicroPython	25078	47035	1.904
SHA-256	C/C++	670.000	701.090	1.046
SHA-256	Rust	711.000	997.320	1.403
SHA-256	MicroPython	596238	1332951	2.236
FFT (Q15)	C/C++	1185.562	1821.500	1.536
FFT (Q15)	Rust	1493.750	3525.500	2.360
FFT (Q15)	MicroPython	241504	844855	3.498
FIR (binary32)	C/C++	188774	30055	0.159
FIR (binary32)	Rust	186340	24219	0.130
FIR (binary32)	MicroPython	4157999	9066768	2.181
IIR (binary32)	C/C++	3268.000	214.320	0.066
IIR (binary32)	Rust	3277.500	248.930	0.076
IIR (binary32)	MicroPython	84702	90164	1.064

6 Reproduction contract

The experimental anchor is commit `933363b342ac960c0d364b4c6379b1ac884dec9f`. C6 ran at fixed/read-back 160 MHz with DFS, sleep, Wi-Fi, BLE, and IEEE 802.15.4 disabled; flash was 4 MB DIO at 80 MHz. One warm-up preceded 100 accepted observations at each of eight frozen sizes. Raw rows retain batch count, total batch duration, normalized duration, checksum, toolchain, vector hash, board, flash, and control metadata where defined by the workload schema.